Year 8
Mathematics

Unit 4: 2D Shapes and 3D Solids

4.6 STEM: Measures of Area and Volume

STEM: Measures of Area and Volume

STARTER · 5 MIN

Date: _____ **Title: STEM: Measures of Area and Volume**

LEARNING OBJECTIVES

- Convert between metric measures for area and volume

STARTER *Answer in your book. No calculators.*

1. How many cm are in 1 m?
2. How many mm are in 1 cm?
3. How many m are in 1 km?
4. Work out 100 × 100

Starter — Answers

STARTER

Tick or correct in green pen

1. **100**
2. **10**
3. **1000**
4. **10 000**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can convert between cm^2, mm^2, m^2 and km^2	☐
SILVER	I can apply area and volume conversions to real problems	☐
GOLD	I can convert between cm^3, mm^3 and m^3	☐

Tick each box in the plenary when you can do it on your own.

Introduction — STEM: Measures of Area and Volume

INTRODUCTION · 3 MIN

1 m = 100 cm, so 1 m^2 = 100 cm × 100 cm = 10 000 cm^2.

Area: cm^2 and mm^2

1 cm^2 = 10 mm × 10 mm

1 cm^2 = 100 mm^2

Area: m^2 and cm^2

1 m^2 = 100 cm × 100 cm

1 m^2 = 10 000 cm^2

Volume: cm^3 and mm^3

1 cm^3 = 10 × 10 × 10 mm^3

1 cm^3 = 1000 mm^3

Volume: m^3 and cm^3

1 m^3 = 100 × 100 × 100 cm^3

1 m^3 = 1 000 000 cm^3

Examples 1 and 2 — Area Units

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint 1 m^2 = 10 000 cm^2 1 cm^2 = 100 mm^2

Example 1

Convert 4.2 m^2 to cm^2.

Example 2

Convert 350 mm^2 to cm^2.

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

1 m² = 10 000 cm²

4.2 m² = 4.2 × 10 000

= 42 000 cm²

Tip: big unit to small unit: multiply.

Example 2

1 cm² = 100 mm²

350 mm² = 350 ÷ 100

= 3.5 cm²

Tip: small unit to big unit: divide.

Task 1

PRACTICE · 9 MIN

BRONZE *Convert*

Hint 1 m^2 = 10 000 cm^2 1 cm^2 = 100 mm^2

1. 3 m^2 to cm^2
2. 5 cm^2 to mm^2
3. 20 000 cm^2 to m^2
4. 800 mm^2 to cm^2
5. 2.5 m^2 to cm^2

★ 0.6 cm^2 to mm^2

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. 30 000 cm²
2. 500 mm²
3. 2 m²
4. 8 cm²
5. 25 000 cm²

★ 60 mm²

Examples 3 and 4 — Volume Units

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint 1 cm^3 = 1000 mm^3 1 m^3 = 1 000 000 cm^3

Example 3

Convert 3 cm^3 to mm^3.

Example 4

Convert 0.01 m^3 to cm^3.

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

1 cm³ = 1000 mm³

3 cm³ = 3 × 1000

= 3000 mm³

Tip: 10 × 10 × 10 = 1000.

Example 4

1 m³ = 1 000 000 cm³

0.01 m³ = 0.01 × 1 000 000

= 10 000 cm³

Tip: 100 × 100 × 100 = one million.

Task 2

PRACTICE · 9 MIN

SILVER *Convert*

Hint 1 cm^3 = 1000 mm^3 1 m^3 = 1 000 000 cm^3

1. 5 cm^3 to mm^3
2. 2 m^3 to cm^3
3. 4000 mm^3 to cm^3
4. 3 500 000 cm^3 to m^3
5. 0.5 m^3 to cm^3

★ 0.25 cm^3 to mm^3

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $5000\ mm^3$
2. $2\ 000\ 000\ cm^3$
3. $4\ cm^3$
4. $3.5\ m^3$
5. $500\ 000\ cm^3$

★ $250\ mm^3$

Examples 5 and 6 — Hectares and Litres

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint 1 hectare = 10 000 m^2 1 m^3 = 1000 litres

Example 5

(3 marks)

1 hectare = 10 000 m^2.

A field is 1.2 km by 1.6 km.

Find its area in hectares.

Example 6

(3 marks)

A tank is 2 m by 1.5 m by 1 m.

How many litres does it hold?

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Change km to m first

area $= 1200 \times 1600$

$= 1\,920\,000\ m^2$

$= 192$ hectares

Tip: divide by 10 000 for hectares.

Example 6

m^3 to cm^3 to litres

$V = 2 \times 1.5 \times 1 = 3\ m^3$

$= 3\,000\,000\ cm^3$

$= 3000$ litres

Tip: 1 m^3 = 1000 litres.

Task 3

PRACTICE · 9 MIN

GOLD *Exam-style: units in real life*

Hint 1 hectare = 10 000 m^2 1 m^3 = 1000 litres

1. Field 400 m by 250 m. Hectares?
2. Park 0.5 km by 0.8 km. Hectares?
3. Tank 1 m by 1 m by 2 m. Litres?
4. Pond 3 m by 2 m by 0.5 m. Litres?
5. Tiles 20 cm by 20 cm. How many cover 1 m^2?

★ Sheet 2 m by 1 m. How many 10 cm by 10 cm squares?

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. **10 hectares**
2. **40 hectares**
3. **2000 litres**
4. **3000 litres**
5. **10 000 ÷ 400 = 25**

★ **20 000 ÷ 100 = 200**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Convert 6.5 m^2 to cm^2

SILVER Convert 7 cm^3 to mm^3

GOLD Convert 0.02 m^3 to cm^3

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **65 000 cm^2**

SILVER **7000 mm^3**

GOLD **20 000 cm^3**

Exam Practice — Area Units

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint 1 m^2 = 10 000 cm^2 1 cm^2 = 100 mm^2

Question 1

(2 marks)

Convert

(a) 8 m^2 to cm^2 (1)

..... cm^2

(b) 650 mm^2 to cm^2 (1)

..... cm^2

Question 2

(2 marks)

A rectangle is 2 m by 3 m.

Work out its area in cm^2.

..... cm^2

Exam Practice — Volume Units

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint 1 cm^3 = 1000 mm^3 1 m^3 = 1 000 000 cm^3

Question 3

(2 marks)

Convert

(a) 9 cm^3 to mm^3 (1)

..... mm^3

(b) 4 500 000 cm^3 to m^3 (1)

..... m^3

Question 4

(3 marks)

A cube has edges of 0.5 m.

Work out its volume in cm^3.

..... cm^3

Exam Practice — Real-Life Units

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint 1 hectare = 10 000 m^2 1 m^3 = 1000 litres

Question 5

(3 marks)

A field is 600 m by 450 m.
1 hectare = 10 000 m^2.

Work out the area of the field in hectares.

..... hectares

Question 6

(4 marks)

A water tank is 1.2 m by 0.8 m by 0.5 m.
Work out how many litres it holds.

..... litres

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) 80 000 cm² B1

(b) 6.5 cm² B1

Tip: m^2 to cm^2: × 10 000.

Question 2

area = 6 m^2 M1

= 60 000 cm^2 A1

Tip: or 200 × 300 in cm.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) 9000 mm³ B1

(b) 4.5 m³ B1

Tip: m³ to cm³: × 1 000 000.

Question 4

0.5 m = 50 cm B1

V = 50 × 50 × 50 M1

= 125 000 cm³ A1

Tip: change to cm first.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

area	$= 600 \times 450$	M1
	$= 270\,000\ m^2$	A1
	$= 27$ hectares	A1

Tip: ÷ 10 000.

Question 6

V	$= 1.2 \times 0.8 \times 0.5$	M1
	$= 0.48\ m^3$	A1
	$= 0.48 \times 1000$	M1
	= 480 litres	A1

Tip: 1 m^3 = 1000 litres.